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Neutral buoyancy trajectories in the ocean

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Abstract

The helix phenomena are one of several advection processes which determine how properties, like heat, salt or nutrients, are vertically transported in the ocean. This phenomenon is a result of studying trajectories that suffer a 0 net force. This happens when the buoyancy force, which results from being in a fluid, and the gravitational force, are equal, and therefore no energy needs to be spent in the movement. In doing so for a closed trajectory for the ocean (in the horizontal plane) it is found that the start and endpoint aren't the same, there's a height difference, a **pitch**. As a result, one can't define a surface for the whole ocean in which every trajectory suffers a 0 net force, but it can be approximated, a technique that's used by oceanographers nowadays.

In this project, we developed a code in Python that computes these helices and their pitch for imaginary closed horizontal trajectories. To do so we make use of a data set of real ocean data points (WOCE) and interpolate between them to get the desired data, as well as the theory of neutral trajectories in the ocean studied by McDougall and Jackett in 1988.

Using our code we've been able of validating the neutral surface approximation. We also checked that our code reproduces real physical phenomena like the Antarctic Polar Front and the Gulf Stream. A lot of work needs to be done to improve this code, but we are happy with this first attempt on it, as it seems to be already a useful tool to determine the difference in height of neutral trajectories and its relationship with thermodynamical properties.

Contents

1	Introduction	2
2	Theoretical background	4
2.1	Buoyancy	4
2.2	Thermodynamic analysis of the ocean	4
2.3	Neutral bouyancy surfaces	4
2.4	Reason behind the helices	5
3	Methods	7
3.1	WOCE Global Hydrographic Climatology	7
3.2	Initial piece of code	7
3.3	Neutral closed trajectories	8
3.4	Sigma 0 calculation	9
3.5	Maps of neutral trajectories and neutral function errors	9
4	Results and discussion	12
4.1	Single trajectories	12
4.2	Maps	13
4.3	Discussion	14
4.3.1	Neutral surface approximation	14
4.3.2	Antarctic polar front	15
4.3.3	Gulf stream	16
5	Conclusions	17

Chapter 1

Introduction

Every particle submerged in a fluid suffers two forces in opposite directions, the gravitational pull and the buoyancy force. In studying the case in which the two forces cancel each other, known as **neutral buoyancy**, we can find a state of equilibrium in the system. If we focus on ocean water parcels as our particles, we can define a surface in which the parcel can make small displacements and still fulfil the neutral buoyancy requirement. We call this the **neutral tangent plane**. If one would follow a parcel doing a closed (horizontal) trajectory that fulfils this principle of no net force, the parcel **won't** end at the same point in height as it started in. We call this phenomenon **helix**, and it's the reason why it's impossible to define a surface in which every trajectory fulfils the neutral buoyancy condition. We say, therefore, that the neutral surfaces are ill-defined, as depicted in figure 1.1.

This helix phenomenon constitutes one of several vertical advection processes, and as so, helps us understand the transport of different ocean properties like heat, nutrients, or salt. It arises from small scale mixing processes as stated by McDougall and Jackett (1988). In the look for more in depth physical reason to this phenomenon, we could only find that explanation, as in their paper McDougall and Jackett focus more on the theoretical aspects of the process than in the actual physical reason. So far we haven't been able to find any other comment about it in literature.

Focusing now on the neutral surfaces, even ill-defined, they are used nowadays to study the behaviour of the ocean in, for example, the equation of state of the ocean TEOS-10 [McDougall & Barker, 2011]. That is because the mixing/displacement of oceanic properties in the horizontal direction is assumed to be in these surfaces, as no work has to be done while staying on them, and the speed at which they displace is way bigger than the vertical one [McDougall, 1987]. That's why (at least in the bigger picture), the vertical advection processes like the one we consider in this report, can be neglected in those cases.

This helix phenomenon in the ocean has already been studied before, [McDougall & Jackett, 1988]. But while, in other studies, the main approach was using mathematical methods to reduce the ill-definition of neutral surfaces, in this project we aim to tackle these helices more computationally. We want to be able to compute these trajectories (using real data), compute the difference in height between the start and the endpoint, and extrapolate that to the whole world map. In doing so we want to develop a code that could be useful to justify the approximation of neutral surfaces, as well as being able of reproducing some real physical phenomena. We will use a data set with over 10000

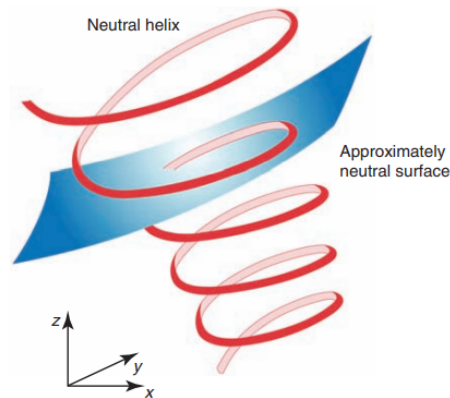


Figure 1.1: Scheme of a neutral buoyancy trajectory and its approximate neutral surface. The neutral surfaces are ill-defined due to the helix phenomena of neutral trajectories, therefore they must be approximate.[McDougall & Jackett, 2009]

vertical profiles of ocean water data points and the programming language Python to achieve this goal. In our approach, we will be looking for general behaviour and its order of magnitude, so, we will ignore precise error calculations as they would have taken useful time from the project, and they are not important to us.

Chapter 2

Theoretical background

2.1 Buoyancy

To begin our understanding of neutral surfaces and therefore neutral buoyancy trajectories we need the concept of buoyancy of a fluid particle given by equation 2.1:

$$b_p \equiv g \left(\frac{\rho_e - \rho_p}{\rho_e} \right). \quad (2.1)$$

This equation gives us a measurement of summing two opposite forces that are being applied on a parcel of fluid (with density ρ_p), surrounded by an environment (with density ρ_e). These two forces are the gravitational pull of the earth being applied on the parcel, and the buoyancy force that emerges from the fluid surrounding the parcel. The sign of b will determine whether the parcel accelerates upwards ($b > 0$) or downwards ($b < 0$).

In setting this buoyancy term to 0 we end up with a neutral buoyancy state (no acceleration). As so, if we look out for trajectories that satisfy this condition at every point we can say we are talking about **neutral buoyancy trajectories**. In doing so we are looking for a trajectory that is a state of equilibrium for the system as the parcel of fluid won't lose any energy across it.

2.2 Thermodynamic analysis of the ocean

As our study will focus on the change of density of the parcel (ρ_p) while submerged in a fluid of another density (ρ_e) we can use an equation of state of the form ($\rho = \rho(p, S, \Theta)$) where the thermodynamic variables (choose by convenience) are:

- Pressure p ,
- Absolute salinity S (g/Kg) [McDougall & Barker, 2011],
- Conservative temperature Θ ($^{\circ}\text{C}$), which is conserved in an adiabatic evolution [McDougall, 2003].

Another two coefficients that will be important to us in defining the evolution of our parcel of fluid are:

$$\beta \equiv \frac{1}{\rho} \left(\frac{\partial \rho}{\partial S} \right)_{\rho, \Theta} \text{ Haline coefficient}, \quad \alpha \equiv -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial \Theta} \right)_{p, S} \text{ Thermal expansion},$$

where β expresses the change of density with salinity and α with conserved temperature.

2.3 Neutral buoyancy surfaces

Focusing now on the neutral buoyancy requirement, that is setting the buoyancy term to 0 (equation 2.1). This sets the two densities ρ_p and ρ_e equal. In doing so we can now invoke thermodynamic

arguments as these two quantities can be expressed by the 3 variables described above ($\rho = \rho(p, S, \Theta)$). We are going to make the next assumptions while doing it:

- The parcel doesn't mix with the environment nor it exchanges any heat ($\delta S_p = \delta \Theta_p = 0$). As we are studying a process that doesn't have to do anything with mixing (in the big scale).
- The change of pressure of the parcel is equal to the change of pressure of the environment ($\delta p_e = \delta p_p = \delta p$)

Taking all this into account we can get to the equation 2.2:

$$\beta \delta S_e = \alpha \delta \Theta_e (\text{neutral buoyancy trajectory ocean}), \quad (2.2)$$

as done in [McDougall & Jackett, 1988]. This equation will be our foundation block for the rest of the project as we will use it to define the neutral buoyancy trajectories we are looking for.

We can now use equation 2.2 to define a *neutral tangent plane* in which a parcel of seawater can be moved by small deviations and still fulfil the neutral buoyancy condition. We could then separate a given trajectory in the horizontal plane in really small steps and attend to the neutral tangent plane in each step to compute which value of pressure assign to each point. In doing so for closed horizontal trajectories we find that we don't end up at the same point in height as we started in, our trajectories don't close in the vertical direction. We have, therefore, **helices** (We call them helices even if they don't follow the geometry of a helix, the name implies the lack of vertical closing).

Therefore neutral buoyancy global surfaces (neutral surfaces) are ill-defined because of this path dependence. As so it will be important for us to understand what's the magnitude of this difference in height for closed trajectories (pitch) so that we can compare it to values like the range of height covered by a trajectory to understand if the neutral surface approximation is valid or not.

2.4 Reason behind the helices

Let's do a mental experiment to understand the helix phenomena, we will perform a neutral trajectory in the shape of a square (in the horizontal plane) as depicted in figure 2.1, as done in [McDougall & Jackett, 2009]. The key point of this path dependence as we will see now is the change of (β/α) with pressure.

In this neutral trajectory, we displace from point A to point B by maintaining the pressure constant, so the evolution is given by the $(\beta/\alpha)(p_A)$ as shown in the thermodynamic space (because of equation 2.2). Then from point B to C, Θ and S are maintained constant, therefore changing the pressure. Therefore, the ratio (β/α) has changed in point C. Because of this when we go from point C to D the path is different from that from A to B, because of this change in the ratio with pressure. In the end from D to E, we maintain Θ and S constant again and end up in a different point in thermodynamic space as we started in, which in position space translates into a difference in pressure, that is, a height difference, the so-called **pitch**.

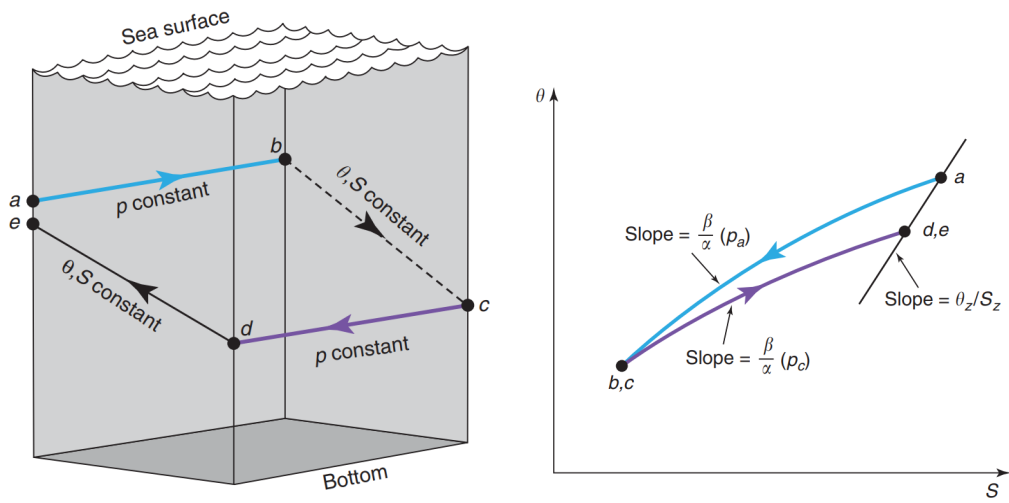


Figure 2.1: Mental experiment to show the path-dependence of neutral buoyancy trajectories in the ocean. [McDougall & Jackett, 2009]

Chapter 3

Methods

The methods used in this project are mainly computational. That is, the database: WOCE Global Hydrographic Climatology [Gouretski & Koltermann, 2004a] and, the programming language, in our case Python. For the context of this project Python is a useful language as it has some libraries specifically made for calculations of thermodynamic parameters in the ocean, like the one we used gsw.

3.1 WOCE Global Hydrographic Climatology

The database we will use is the WOCE Global Hydrographic Climatology, It was published in 2004 and It has around 10000 vertical profiles (casts) distributed across a $0.5^\circ \times 0.5^\circ$ grid for the whole planet. In each cast there are data points distributed in this standard levels: *0, 10, 20, 30, 40, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 1000, 1100, 1200, 1300, 1400, 1500, 1750, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750, 4000, 4250, 4500, 4750, 5000, 5250, 5500, 5750, 6000 meters.*

As noted by the paper [McDougall & Jackett, 1988], the pitch of helices found in the North Atlantic is around the order of 10 m. If we compare this pitch with the difference in height of our available set of data points, It's clear that we will have to resort to interpolation using the data points as a reference to get intermediate values of pressure, salinity and conserved temperature, to do so we will use a parabola interpolation with the data points. It's important to note that as we go deeper the resolution of our data reduces by 25 times, therefore the data obtained for deeper oceans will have a bigger relative error assigned to it.

3.2 Initial piece of code

The programming language used (Python) was given to us from the beginning since the initial function that we will use throughout the project was written by our supervisor Czaja, Arnaud in the same language. Thus, that initial function, which from now on I will call **neutral function**, was our foundation block for the code.

This function will take us from a point A of known position in physical and thermodynamical space to a point B of known longitude and latitude while following the neutral trajectory a parcel of water would follow. The initial parameters needed are: the latitude, longitude, height, conservative temperature and salinity of the start point A. And the longitude and latitude of the point B we want to go, as well as values from data points of the vertical cast in B to interpolate with them.

We will assume that the height of point B will land at a similar height that to A (as we assume the points are close together), this is, we will take N data points above and below (N_{abv}) the starting height of A (Having $2N_{abv}+1$ data points). Being precise what we mean with the starting height of A is the nearest data point in height to A. Inputting all of this data into the neutral function, we will get back the height, salinity and temperature of our point B (that don't need to land on a data point), as depicted in purple in figure 3.1. To do this, the function made a fit of the data points to a quadratic function and then solves the neutral buoyancy equation 2.1.

Datapoints

Closest datapoint to A

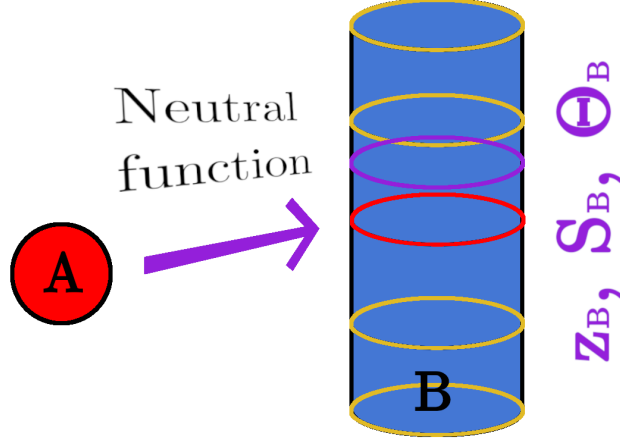


Figure 3.1: Scheme of the functioning of the neutral function used. The input values of the function are height (z), pressure (p) and salinity (S) of point A and the data points inside the blue cylinder. This cylinder is defined by the coordinates of point B and the number of data points we want to consider. This is defined by another parameter (N above and below, N_{abv}). In this figure, N_{abv} is 2 (5 data points in total) which are the yellow circles plus the red one, as that is the closest data point to A in cast B. In calling the function it will return the values marked with purple of height, salinity, and conservative temperature of point B that a neutral trajectory would follow if it went from point A (of known starting height) to a point in the horizontal plane B (of unknown height).

3.3 Neutral closed trajectories

Using now the neutral function in a closed-loop the desired neutral closed trajectories can be obtained. In our code, we can do two different types of horizontal trajectories, ellipses or rectangles, as shown in figure 3.2

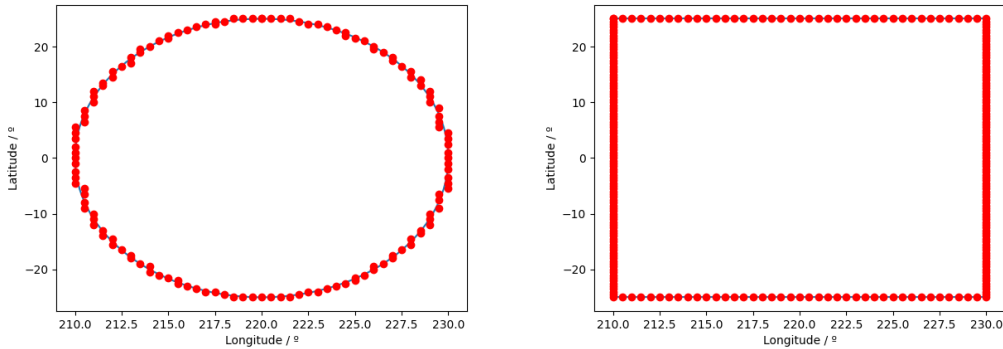


Figure 3.2: Possible computational horizontal trajectories in our code. The red dots represent the actual casts from the data set the trajectory will go through. We can change both R_x and R_y of the ellipse (where x will be longitude and y latitude direction), as well as both sides of the rectangular shape. We can also change the coordinates of the centre of the ellipse and the rectangle.

In choosing the dimensions of the trajectories we will prioritise bigger shapes, this is because we want to magnify the path dependence, and as a result, the pitch obtained. As so we will prioritise bigger shapes in the vertical direction, as that's the way the oceans are disturbed. Therefore getting more possible neutral trajectories than if it were otherwise.

With the neutral function and closed horizontal trajectories, we can start computing neutral buoyancy trajectories by defining a start point in height, for a given horizontal trajectory, and from there use the neutral function in between two consecutive casts (determined by the horizontal trajectory) to get the height of the next point. In doing these across all the casts of the trajectory we get the desired neutral buoyancy trajectories.

3.4 Sigma 0 calculation

In all the above calculations we have used the coefficients α and β with the in-situ values of pressure, salinity and conservative temperature. But, there is another parameter that we can change in the neutral function that changes the way these coefficients are calculated. This method is that rather than using the in-situ values α and β , it calculates them at pressure 0 (what we call σ_0). In this way, we eliminate the pressure dependence that was key in the neutral buoyancy trajectories to obtain the pitch.

Thus, we would expect the pitch of these σ_0 calculations to be 0. In reality, we do not observe this, but, we can assign the values obtained by these calculations as computational errors. In other words, we are going to use the σ_0 calculation as a way of estimating errors. These errors could come from many different sources, as our data is transformed and manipulated many times. The main reason we can assume this error comes from is the interpolation performed by the neutral function, because as we saw, the data set has a really limited resolution, particularly at the deepest heights. It could come as well from physical processes we are not taking into account. I will come back to this point in the results section.

3.5 Maps of neutral trajectories and neutral function errors

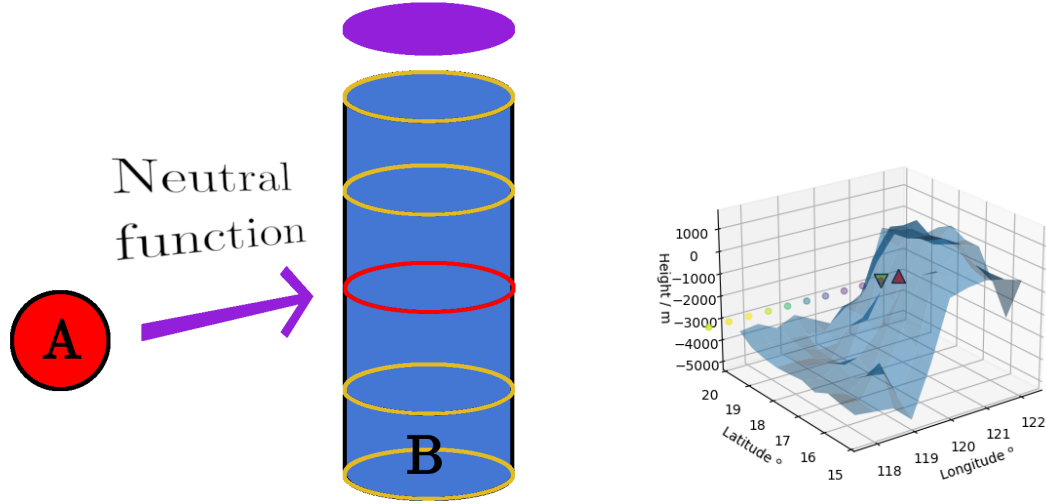
The neutral trajectory code described above can now be used iteratively in the world map to find the general behaviour of our studied phenomenon. To do so we make a grid on the world map and at each point, we will compute a neutral trajectory using that point as a centre. Ideally one would like to choose the separation between each calculation to be 0.5° , as it is the resolution of our data set. In reality that would translate into a lot of computational time just to get one map, in the order of 10s of hours.

As so we are dividing our grid by a smaller factor. From now we will use a separation of 4° for the longitude direction, and a separation of 2° for the latitude. We found this combination was the best one in terms of getting a good resolution and reducing computational time. With this, our maps take approximate 3 hours to get done, which, still, makes getting a good amount of data a harsh job. More work could be done to reduce the computational time but that would imply rewriting the entire code to fit it in a multi-processing or multi-threading python library, something like that is way beyond our abilities and the scope of the project. Although we did try our best to reduce the computational times by optimising the code in every possible way.

To this point the distribution of work between my partner and I was 50/50 when it came to the code. By this moment in the project the computer of my partner wasn't able of computing the next results so we distributed the work in a different way, from now on he focused more on the theoretical aspects and the literature and me in the actual coding and obtaining results.

After these considerations, we found that the maps did were missing a lot of points that we couldn't explain. To understand them we need to study the neutral function built by our supervisor and check the type of errors there are. In doing so we find 3 types of errors:

- **Crit neg** error, the function couldn't find a solution for the quadratic equation, as shown schematically in figure 3.3 a).
- **N2 neg**, the function found a solution, but it's physically impossible. I.e. negative square roots.
- **No data**, we inputted a vector with empty values into the function. Shown in figure 3.3 b).



(a) Crit neg error . It happens when the code couldn't find a solution (purple), because it is outside of our considered range (blue).

(b) No data error. The blue surface is the bottom of the ocean. In this case the trajectory encountered the Philippines islands at about 120°E 12.5°N.

Figure 3.3: Two different schemes of the kind of errors obtained by the neutral function used.

With the errors categorised we can now use one of our map calculations and plot in each point we get an error the type we got. In doing so we get figure 3.4 (This plot is done for the neutral calculation and not σ_0 but both share the same error points apart from some minor differences). We can see how the no data points (blue) follow the shape of the continents, as expected, we can't have ocean measurements on the land. The crit neg (green) points are a special physical phenomenon in which some high steeped neutral surfaces appear. I will explain this more in detail in the discussion section.

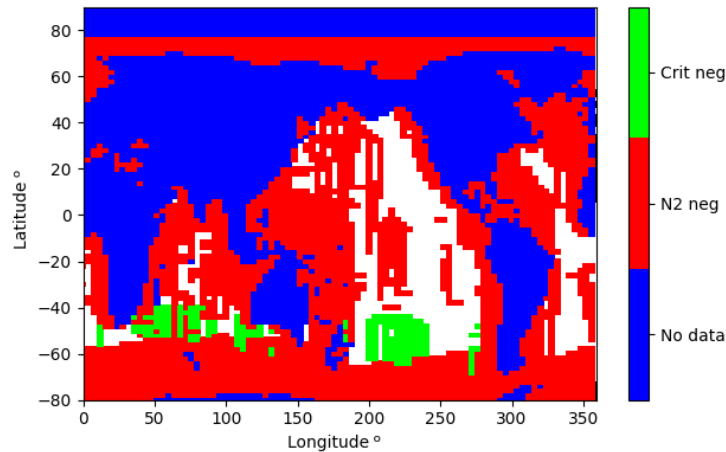


Figure 3.4: Errors of neutral buoyancy trajectories computed with our code. Each square represents the error code of a trajectory to which that point was the centre. In this map the horizontal trajectory used was a rectangle of $side_x = 10^\circ$ and $side_y = 25^\circ$. The initial height of every trajectory was 909.0 m.

In studying the errors N2 neg (red), we didn't find an actual reason why they should be appearing so often. So, we decided to check if the N2 neg errors were coming from points that were below the bottom of the ocean, although those errors were supposed to be no data errors. In doing so, we found that, in fact, the N2 neg errors were no data errors all along. This is because of the way the data set is built, due to some data points which are below the bottom of the ocean, being non-empty. They are accessible but **masked**. What masked means (in the context of python) is that the data is in the array, but, non-accessible (there's a "mask" on top of it) by other functions. For reasons that escape the scope of this report these masked values weren't being recognised as an error at first, and later when doing the calculations needed, the function encountered the N2 error as a result.

Chapter 4

Results and discussion

4.1 Single trajectories

Using our neutral function to get the desired neutral trajectories, we obtain the figures 4.1 (for 2 arbitrary horizontal trajectories). From these figures we can't extract much relevant information for our study as they are quite arbitrary. Therefore we are going to extrapolate this code to the world map. But before that, we can use this plots to understand how our code behaves.

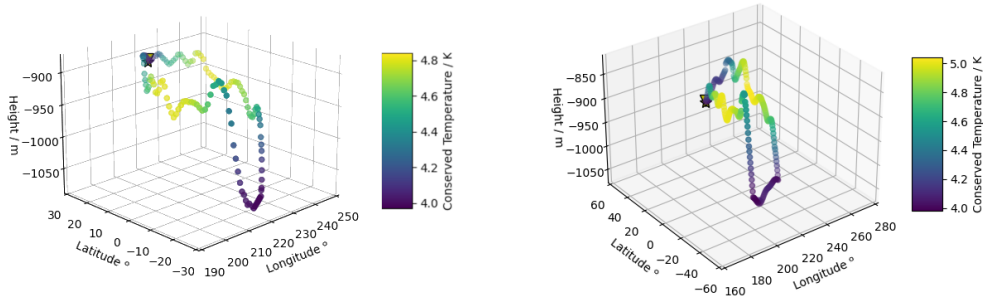


Figure 4.1: Two neutral buoyancy trajectories done with our code. On the left the horizontal trajectory is an ellipse of $R_x = 10^\circ$ and $R_y = 25^\circ$, and on the right a square of side $_x = 20^\circ$ and side $_y = 50^\circ$, both are centred in 40°W and 0°N . The star marks the starting point of the trajectory and the triangle at the end, although both overlap because of the vertical range so they are hard to see. The pitch found for the left graph was -0.2 (-4.8 m (neutral calculation) - (-4.6) (σ_0 calculation)) and for the right -1.1 m (-2.2 m (neutral calculation) - (-0.9) (σ_0 calculation)). The negative sign implies the trajectory ends at a point below the starting one.

The only surprising result we can extract from doing some neutral trajectories with this piece of code is that the σ_0 calculation is not giving us the pitch 0 expected, instead, we are getting results that are quite close to the neutral calculation. As we said before, we were going to use this σ_0 method as accounting for the error, so, this non zero result could be leading us to some physics we are not taking into account. This could be either, that, not only the pressure dependence of β/α is key to the helix phenomena, or, that we are getting more vertical advection processes mixed in our results. As of today, we haven't been able of pointing out which of those two is, although the more reasonable answer is the latter. Because the helix phenomenon has been known for a long time now, so it seems unreasonable to assume there are new physics behind such a simple calculation. And as mentioned before in the methods chapter, it could also be a computational error coming from all of the different transformations our data goes through.

4.2 Maps

Using the code described in the last section, we can now use it iteratively in the whole world map to get figures like 4.2. We can make this map now for different starting heights. As shown in figure 4.3. In doing so we can appreciate how, as we go to deeper heights, the amount of points get reduced.

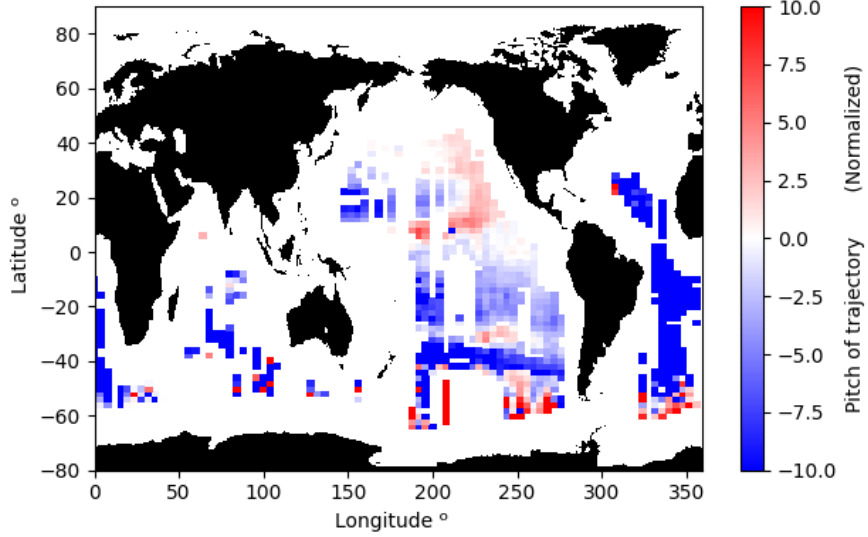


Figure 4.2: Pitch of neutral buoyancy trajectories. Each square represents the pitch of a trajectory to which that point was the centre. In this map the horizontal trajectory used was a rectangle of $side_x = 10^\circ$ and $side_y = 25^\circ$. The initial height of every trajectory was 909.0 m. And we get an average pitch of -9.9m (without normalisation). The pitches have been normalised using the built-in function of the python library matplotlib, this is a linear normalisation in which the maximum is taken as 10 and the minimum as -10. We chose this normalisation due to the different range of pitches observed that made the visualisation of some parts impossible.

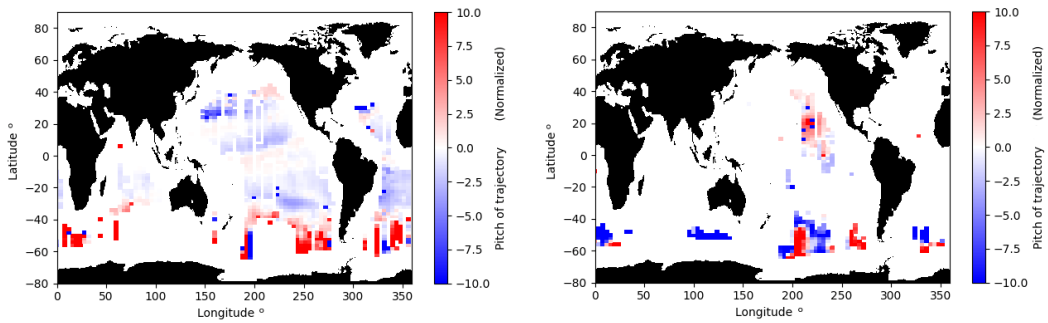


Figure 4.3: Pitch of neutral buoyancy trajectories. The parameters chosen for the trajectories are the same as in figure 4.2. For the left map, the starting height is 404.0 m and the average pitch found is -2.4 m. For the right map, the starting height is 1416.0 m, and the average pitch is -2.4 m. When comparing these two maps and figure 4.2 it can be appreciated that as we go to deeper heights we lose more points. This is due to our trajectory encountering the bottom of the ocean, and, therefore, not being able to compute it.

Other quantities we can plot in this kind of map are the range of values a trajectory goes by. This is the maximum minus the minimum value of a certain quantity we want to study. In our case we did it with Θ , S and height, getting the plots in figure 4.4. We can observe in plot c) a high increase in the range of salinity in the Atlantic Ocean, this relates to the Atlantic Ocean being the ocean basin with the most salt out of the 5 [Gouretski & Koltermann, 2004b]. Therefore, we can already observe our code reproducing real physical phenomena.

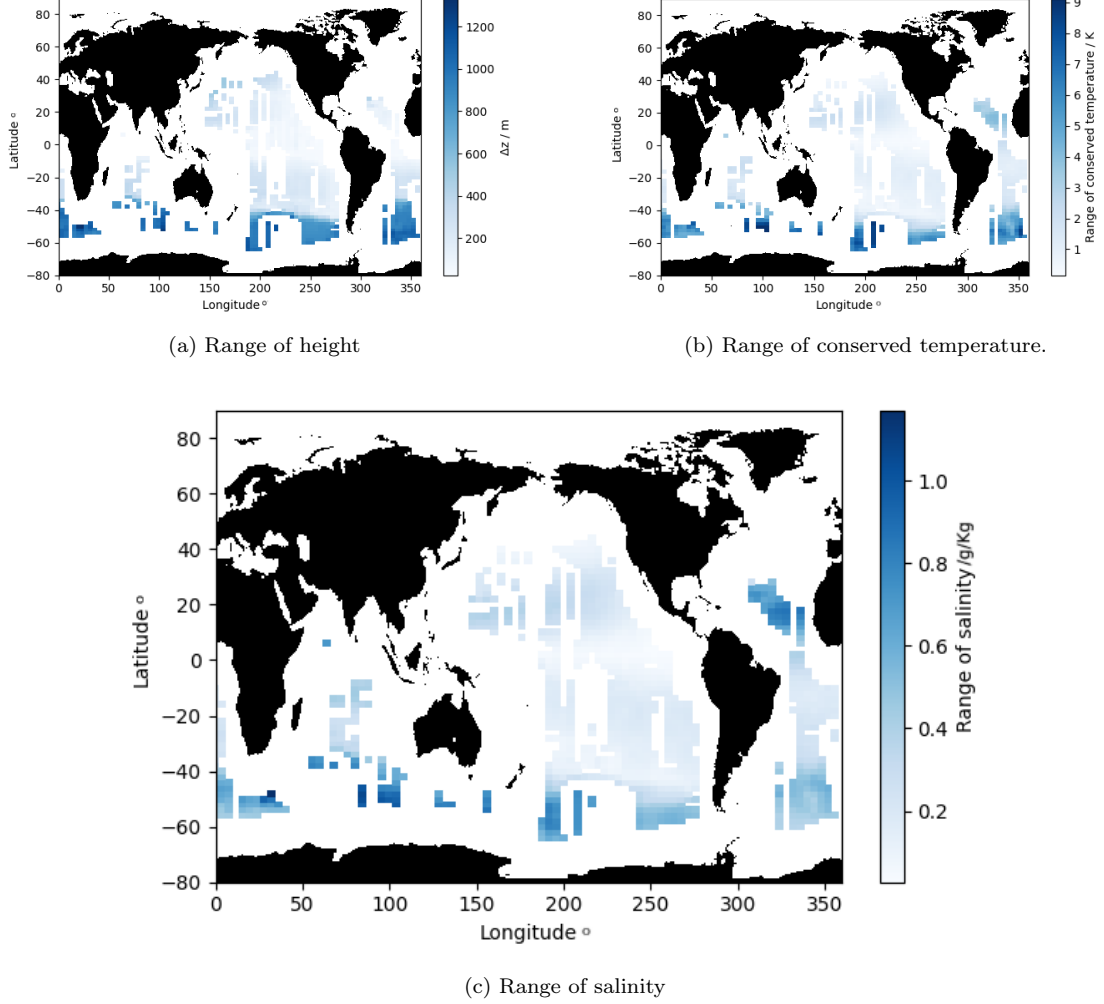


Figure 4.4: Maps of ranges of different quantities (maximum value - minimum) that neutral trajectories go through. Each square represents the centre of a neutral trajectory with the same parameters as the figure 4.2.

4.3 Discussion

4.3.1 Neutral surface approximation

As we've seen in figure 4.2 the average of the pitch is about 10 m (in absolute value). We can now compare this value with figure 4.4 a). In doing so we can check that the order of magnitude of the range of heights is around the hundreds of meters, rounding the 400-600 m in most oceans. Therefore, we can use this difference of order of magnitude to justify the use of approximate neutral surfaces.

Studying the maps from figure 4.3 we get to the same conclusion. This was an expected result as, from what we've seen from literature [Jackett & McDougall, 1997], approximate neutral surfaces are a total valid approximation and are used nowadays in oceanography.

4.3.2 Antarctic polar front

When studying the errors of our code in the 3.5 section, we said we were going to delay the explanation of the green points (the crit neg errors) of figure 3.4. Now if we look at figures 4.3, in particular, figure a), we can see how in the South Ocean It seems the trajectories are travelling way more vertical distance than in other parts of the ocean, and as so the range of Θ and S increases. We can relate this effect with the crit neg errors mentioned before, the trajectories in that region are getting so stepped that to go from one point to the next one the solution lands outside of our considered range.

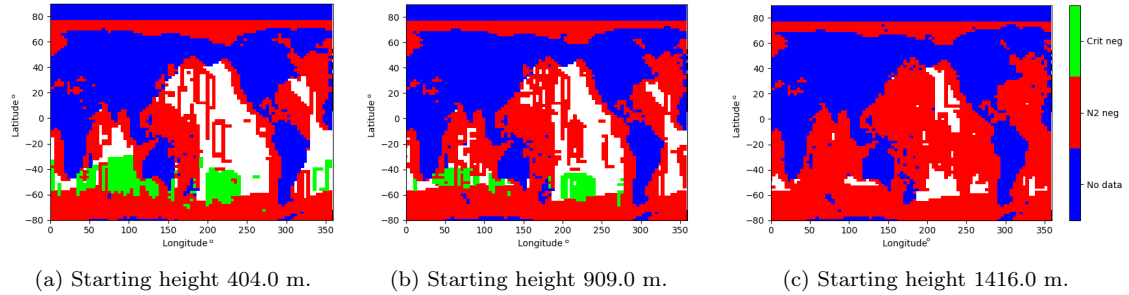


Figure 4.5: Error map done in the same way as 3.4 In this map the horizontal trajectory used was a rectangle of $side_x = 10^\circ$ and $side_y = 25^\circ$.

We can now relate both of these effects in our figures to a phenomenon that surrounds Antarctica in which, northward-flowing water coming from Antarctica encounters warm waters coming from higher latitudes. As so, the cold waters sink abruptly below the warm waters, forming some high steeped neutral surfaces. This phenomenon is called Antarctic Polar Front and is situated around the Antarctic between latitudes 65° S and 45° S [Moore *et al.*, 1999]. This is what we are seeing in our plots.

We could estimate at which height this tongue of cold water stops sinking and starts mixing with the water. To do so we are going to look at 3 different error map plots in figure 4.5 at 3 different heights. A trend of green points disappearing is observed as we increase the starting height, i.e. we start our trajectories deeper. Using this we can estimate that, as we observe no crit neg errors in the 1416.0 m plot but yes on the 909.0 m we can deduce this "tongue" of cold water must stop in that range of heights. To confirm it we will attend to our database and plot a profile temperature for Antarctica to see approximately where this effect ends, this is shown in figure 4.6 and confirms our assumptions, the abrupt steep of cold water ends at around 1000 m depth.

We are conscious of this not being the most precise method of determining the location of this phenomenon. We wanted to illustrate how we could use our code to depict real known physical phenomena, which validates it as a useful tool. To make this calculation more precise we would need to do more maps and study the tendency that the crit neg errors follow. While valid, these claims are outside the scope of our project.

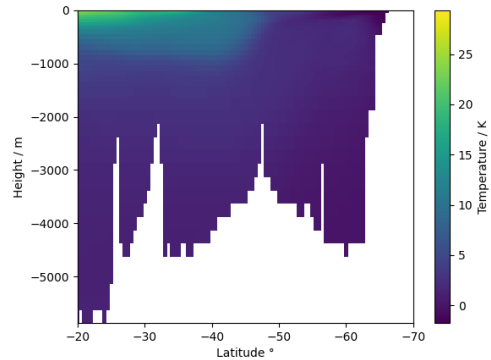


Figure 4.6: Vertical profile temperature at longitude 100° E. The Antarctic polar front can be observed between latitudes 40° S and 50° S. It's observed how the effect ends at approximately 1000 m depth.

4.3.3 Gulf stream

When studying the 4.4 (b,c) a region that stands out is the North Atlantic (20° to 40° North). It can be seen how conservative temperature and salinity cover a bigger range in that region. As so we can identify this with another front of cold water encountering warm waters, this front is related to the Gulf Stream, which forms part of the Atlantic Meridional Overturning Circulation (AMOC) [Joyce & Zhang, 2010].

Again like with the Antarctic polar front, as the cold water sinks underneath the warm waters, the neutral surfaces get really stepped. As so, we see that reflected in our maps. The importance of the Gulf Stream can be appreciated as, the high range of temperature we can see points that this current of water is responsible for a huge heat transport in this ocean [Hofmann & Rahmstorf, 2009].

Chapter 5

Conclusions

We developed a successful code that helps us understand better the helix phenomena, as well as its relationship with the neutral surfaces on the ocean. Besides observing some real ocean phenomena, which validates our code to some extent. As we weren't looking for exact results or data we can't give any concrete numbers about anything in particular as there weren't any precise error calculations at all. With what we've built we can only do estimates of quantities, which is already a success for us.

One useful use we think our code might have is determining when is best to use a neutral surface. What we mean with this is comparing the pitch obtained in a certain oceanic region and then comparing it with the assumed magnitude of the physical phenomenon that wants to be studied, and then, decide if it's better to use them or use another approach.

In terms of further work, there are a lot of open ways that this project could go. Firstly it's important to consider that throughout this project we did imaginary horizontal trajectories, in reality, these closed trajectories don't occur. To account for real currents we would have to take real data and implement it in our code to study how are these currents affected by the helix phenomena.

We would have liked to have more data (and results) to present in this project but the amount of time it takes to just form 3 maps (10 h approximately) made it impossible for us. The code would need to be rewritten to optimise it, and even then it would take some relevant time to get this data. Some effects we wanted to study but we didn't have much time left to do so that could be a good starting point for further work are:

- the effect the range parameter has on our pitch values,
- what do the direction the trajectories imply in terms of oceanic properties,
- how are the pitch and direction of the trajectories related in different depths,
- how the coefficients α and β evolve along the trajectories as well as its derivatives with pressure.

The list could go on and on as this is a quite new concept for us as well as the programming language so we couldn't tackle much in the time available.

Ideally one could implement this code (or at least the idea of it), to the equation of the state of the ocean. Therefore implementing the path dependence of neutral buoyancy trajectories to it. In practice, we assume this might be impossible as the computational power to do maps like 4.2 is high enough, assuming the simplicity of it, that nowadays it's non-viable, but could be a good goal for the close future.

One idea we had since the beginning (but didn't have time to implement as it was too ambitious) was, instead of doing fixed closed horizontal trajectories, using some type of random step algorithm iteratively around the same point. With this we could study which direction a neutral trajectory is most likely to head, and relate it to the actual ocean currents to see to which degree the helix phenomena affects them.

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